ACME: Adaptive Cognitive Memory Engine

Externalizing Belief, Memory, and Learning from LLM Weights

A production-tested cognitive memory substrate for belief lifecycle management in LLM agents.

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LLM □ Episodic Memory □ Belief Engine □ Feedback Loop □ Updated Beliefs

Code: `https://github.com/KamilBourouiba/ACME` (tag `v0.1.1-arxiv`)
Project site: `https://kamilbourouiba.github.io/ACME/`
Data: `GET /api/v1/benchmark/export` (Postgres `benchmark_runs`)

Abstract

Large language models lack durable episodic memory, explicit belief states, structured forgetting, and mechanisms to learn from failure [1,2]. Recent systems report that agents cannot epistemically separate observation from belief [50] and that intrinsic epistemic agency *— including belief updating — remains weak in base models [51]. We present* ***ACME*** *(Adaptive Cognitive Memory Engine), an external cognitive substrate that treats the LLM as a language processor while delegating persistence, confidence tracking, contradiction handling, and self-correction to specialized engines. ACME operationalizes:* when does a collection of experiences deserve to become a belief? *We contribute (1) a CRS-governed belief lifecycle inspired by cognitive architecture [3,4] and truth-maintenance systems [21,54]; (2) hybrid graph + pgvector retrieval [5,6] with contrarian verification [7,8]; (3)* ***MemoryBench v3.1*** *— fourteen sandbox-isolated scenarios with RAG [9], MemGPT [10], and LangGraph-style [11] baselines scored by an LLM judge [12,13]. On Azure OpenAI GPT-4.1, ACME achieves overall* ***0.925*** *vs* ***0.487*** *for vector-RAG (job* `3b31e5e3`*), with feedback and belief-quality metrics unavailable to baselines. Model-sensitivity runs on GPT-4.1-mini and GPT-5.4 preserve a* ***+0.39–0.44*** *margin over RAG. Production ablations show belief sync is essential (−0.19 overall when disabled).*

1. Introduction

Retrieval-augmented generation (RAG) [9] augments LLMs with document search but does not maintain explicit epistemic states, handle contradictory evidence, or learn from prediction failures. LongMemEval [37] shows commercial assistants and long-context LLMs suffer **30–60%** accuracy drops on sustained interactive memory versus oracle evidence — retrieval alone is insufficient. Hindsight [50] argues current agent memory cannot distinguish *what was observed* from *what is believed*, nor maintain preference-consistent reasoning across sessions. Reflection-Bench [51] identifies **belief updating** and meta-reflection as core yet underdeveloped dimensions of epistemic agency in LLMs.

Dense and hybrid retrievers [14,15] improve recall yet remain stateless: they neither promote stable beliefs nor demote refuted hypotheses. Agent memory frameworks — MemGPT [10], generative agents [16], Zep [20], and LangGraph-style orchestration [11] — extend context windows via paging, temporal graphs, or state graphs but lack auditable belief lifecycles and structured feedback loops comparable to cognitive architectures [3,17].

Episodic and semantic memory have long been distinguished in psychology and AI [18,19]. Recent work on long-term conversational memory (MemGPT, memory streams [16], Zep [20]) focuses on *what to store and retrieve*, not *when accumulated evidence warrants belief*. ACME closes this gap with an explicit promotion pipeline (Observation -> Inference -> Hypothesis -> Belief) and symmetric demotion under contradiction — aligning with truth-maintenance and belief-revision traditions [21,22].

Contributions

1. A CRS-governed belief lifecycle with promotion and demotion under contradiction.
2. A hybrid graph + vector retrieval substrate (PostgreSQL/pgvector + Neo4j).
3. Contrarian verification for high-confidence answers.
4. MemoryBench v3.1 with scored belief and feedback metrics.
5. Production benchmark evidence across GPT-4.1, GPT-4.1-mini, and GPT-5.4.

Thesis. RAG optimizes recall; ACME optimizes epistemic state—when experiences become beliefs, how contradictions demote them, and how outcomes close the prediction loop. Our claim is not universal SOTA on every long-context benchmark [37,38], but the first deployed system with sandbox-isolated evaluation of **feedback correction** and **belief quality (CRS)** against RAG/MemGPT/LangGraph under reproducible production runs [49].

2. Related Work

Retrieval-augmented generation. Lewis et al. [9] established the retrieve-then-generate paradigm for knowledge-intensive tasks. Subsequent systems add re-ranking [14], query rewriting [27], and self-critique — Self-RAG [28] and Corrective RAG [29] route or revise retrieval based on generation quality. ACME differs by persisting structured beliefs and failure traces outside the LLM, not only refining a single-turn retrieval pass.

Graph-augmented memory. Knowledge-graph RAG [30,31] and GraphRAG [32] extract communities and summaries from document graphs. ACME maintains a *live* typed graph (entities, causal edges, tenant scope) updated on every experience, coupled with vector recall over episodic embeddings — hybrid retrieval akin to [5,6] but with belief-weighted context assembly.

Agent memory systems. MemGPT [10] virtualizes context via OS-inspired paging; generative agents [16] use memory streams and reflection; LangGraph [11] composes stateful agent workflows. These systems improve *capacity* and *session continuity*; ACME adds CRS-governed belief promotion, contradiction logging, and prediction-outcome loops absent from reference baselines in our evaluation.

Learning from feedback. Reflexion [33] and ExpeL [34] use verbal reinforcement from trial outcomes; Constitutional AI [35] shapes behavior via principles. ACME's feedback engine ties outcome signals to specific graph-backed beliefs, adjusting confidence and status (e.g., Challenged -> Deprecated) rather than only updating prompt-level reflections.

Cognitive architectures. ACT-R [3] and SOAR [17] model declarative memory, production rules, and learning from impasse. ACME is not a full cognitive architecture but borrows the separation of *fast reasoning* (LLM) from *slow accumulative memory* (Postgres, Neo4j, belief records) — similar in spirit to dual-process accounts [4] and complementary learning systems [36].

Evaluation of memory systems. Benchmarks such as LongMemEval [37], LoCoMo [38], MemoryBank [39], MemoryAgentBench [52], and MemBench [53] stress long-horizon recall, test-time learning, or reflective memory. Reflection-Bench [51] evaluates belief updating as one of seven epistemic dimensions. None jointly measure sandbox-isolated **feedback correction**, **CRS belief quality**, and **contrarian groundedness** in a production-deployable API. MemoryBench v3.1 fills this gap (Table 1).

Belief-first and epistemic agent memory (concurrent work). TruthKeeper [54] applies dependency-aware truth maintenance to LLM memory. MnemeBrain [55] stores beliefs with Belnap four-valued logic and AGM-style revision. NeuSymMS [56] couples neural extraction with symbolic TMS via rule engines. Graph-native cognitive memory architectures [57] formalize versioned belief revision for auditability. Hindsight [50] achieves strong LongMemEval/LoCoMo scores via retain-recall-reflect tiers but does not expose CRS-governed promotion thresholds or prediction-outcome loops. **ACME** differentiates by (i) integrated seven-engine cognitive loop with production deployment, (ii) MemoryBench v3.1 scoring belief/feedback dimensions absent from retrieval-centric baselines, and (iii) persisted `benchmark_runs` for third-party reproduction.

3. System Design

Ingestion: LLM extraction [40] plus deterministic normalization -> Neo4j entities/relations + episodic embeddings [23].

Query: Hybrid retrieval (graph neighborhood [24,30] + pgvector cosine [5]) -> LLM reasoning -> optional contrarian pass [7,8] when confidence >= 0.8.

Feedback: Outcome signals update beliefs [21], log failures [33], trigger consolidation (scheduled consolidation, 6 h).

CRS (Cognitive Reliability Score) = 40% prediction success + 20% temporal stability + 20% contradiction resistance + 20% source diversity — a composite reliability index analogous to calibrated confidence under heterogeneous evidence [41,42].

Causal edge types: `observed_with`, `precedes`, `correlates`, `causes`, `disproves` — distinguishing correlation from causation per Pearlian and epistemic traditions [43,44].

Multi-tenancy: Postgres rows and Neo4j nodes keyed by `tenant_id` (header `X-Tenant-ID`).

Ablation toggles: `ABLATION_DISABLE_CONTRARIAN`, `ABLATION_DISABLE_BELIEF_SYNC`, `ABLATION_DISABLE_VECTOR` (see Results).

4. MemoryBench v3.1

Four metrics, LLM-as-judge [12,13] (semantic; keyword overlap reported separately):

Table 1: MemoryBench v3.1 metrics.

Metric	Description
Retention	Concept coverage (synonyms accepted)
Groundedness	Answer supported by ingested episodes
Feedback correction	Belief adjustment after contradictions
Belief quality	Mean CRS across tracked beliefs

Overall = average of all four. Baselines score 0 on feedback/belief (N/A), capping baseline overall near 0.48–0.49.

Fourteen scenarios: retention, contradiction, multi-source conflict, error injection, **knowledge update** (LongMemEval KU [37]), long-term retention, hallucination resistance, feedback adjustment, adversarial noise, long-horizon noise, tenant isolation, healthcare domain transfer, multi-session recall, prediction-outcome loop.

Isolation: Each scenario deletes prior benchmark-tagged Postgres rows and Neo4j subgraph before run (`acme/evaluation/sandbox.py`) — preventing cross-scenario contamination noted as a failure mode in long-horizon evals [37,38,51].

Baselines: Minimal reproducible implementations aligned with Lewis et al. [9] (RAG), Packer et

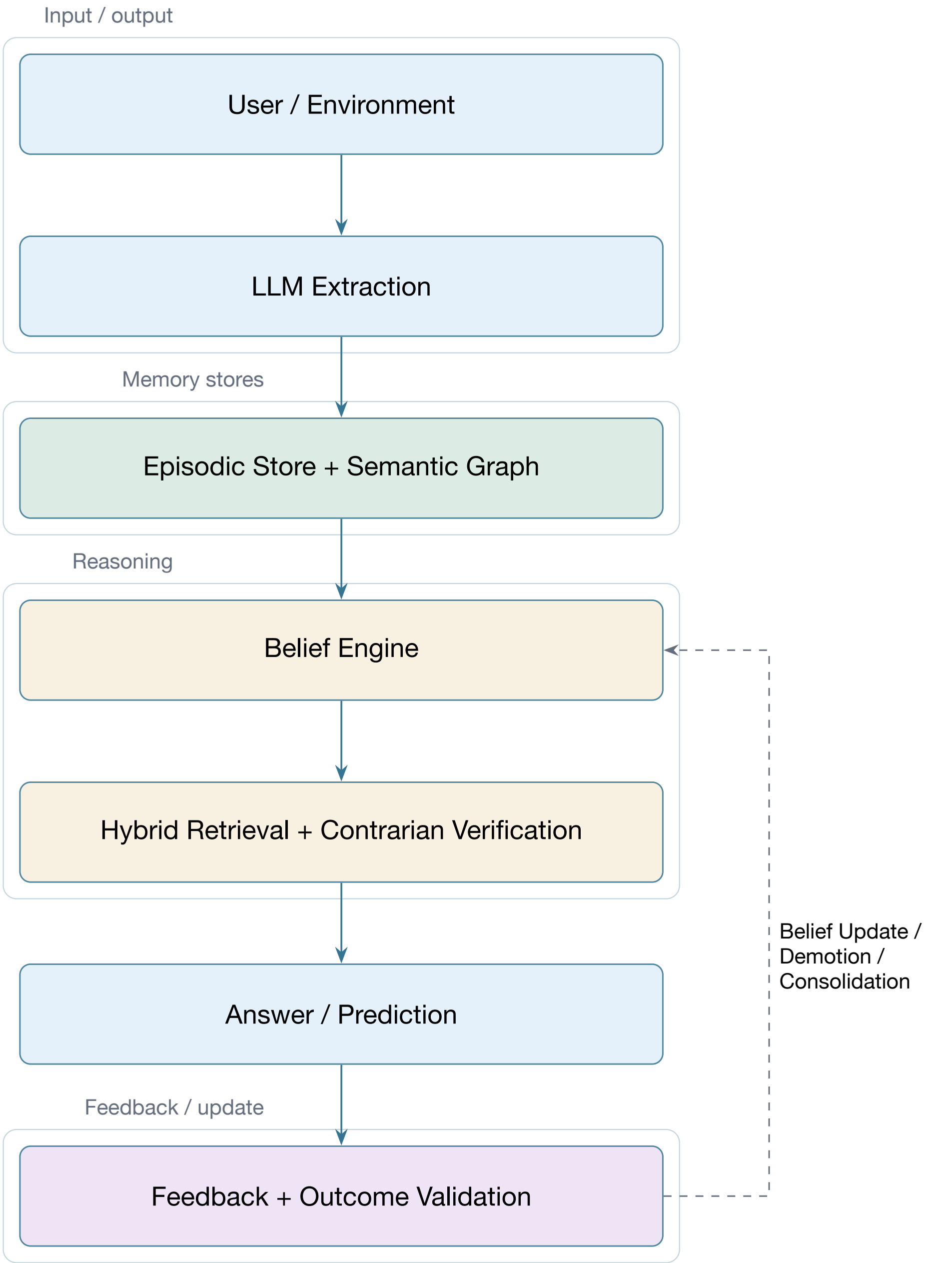

Figure 1: ACME cognitive loop: extraction, hybrid memory, belief governance, and closed-loop correction.

al. [10] (MemGPT), and LangGraph-style state graphs [11] — see `docs/BASELINES.md`.

Primary reported results use the **13-scenario v3** suite (job `3b31e5e3`, pre-`knowledge_update`) for strict comparability; v3.1 adds the LongMemEval-aligned knowledge-update probe.

> **Key finding.** MemoryBench v3.1 is the only evaluated suite in our comparison that jointly scores **feedback correction**, **CRS belief quality**, and **contrarian groundedness** under per-scenario sandbox isolation with a production API.

5. LongMemEval (industry benchmark)

To complement MemoryBench, we integrate the official **LongMemEval** oracle split [37] (500 questions, evidence sessions only). Chat histories are ingested via the same production orchestrator; answers are scored with the **official LongMemEval yes/no judge prompts** from the reference implementation.

Protocol. Each question resets sandbox state (`longmemeval` tag). Haystack sessions are serialized as multi-turn user/assistant transcripts and ingested as experiences. ACME uses a **transcript-first** answer path: sessions are ranked newest-

Table 2: Benchmark positioning — evaluation capabilities (Table 1a).

Capability	LongMem Eval	MemAgent Bench	Reflection Bench	Mem Bench	Memory Bench
Long-horizon / multi-session	Yes	Yes	Partial	Yes	Yes
Knowledge update	Yes	Yes	Partial	Partial	Yes
Feedback / belief revision	No	Partial	Yes	Partial	**Scored**
Hallucination / abstention	Yes	Partial	Partial	Partial	Yes

Table 3: Benchmark positioning — infrastructure and belief metrics (Table 1b).

Property	LongMem Eval	MemAgent Bench	Reflection Bench	Mem Bench	Memory Bench
Per-scenario sandbox	No	Partial	Yes	Yes	**Yes**
Contrarian verification	No	No	Partial	No	**Yes**
Belief quality (CRS)	No	No	No	Partial	**Yes**
Production API + persisted runs	No	No	No	No	**Yes**

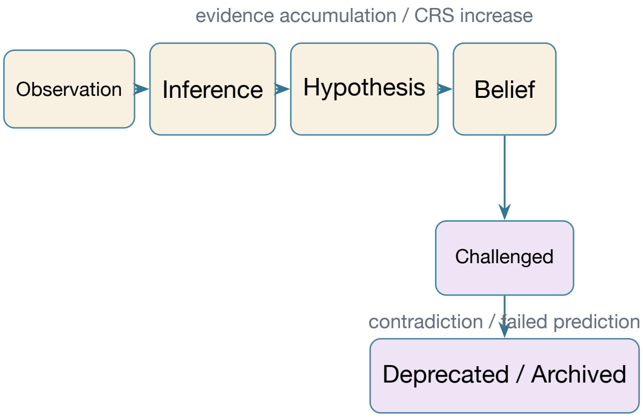

Figure 2: Belief lifecycle: promotion along evidence accumulation and demotion under contradiction.

first, the LLM reads full chat transcripts (with belief graph as secondary context), and knowledge-update items trigger belief demotion on superseded sources. RAG and MemGPT baselines answer from retrieved context; an independent LLM judge (same deployment family as MemoryBench) applies type-specific rubrics (`knowledge-update`, `temporal-reasoning`, `abstention`, etc.).

Production run (June 2026): job `45623ca0`, image `acme-api:longmemeval-v5-hybrid`, **500 Q clean** (~6.2 h). Summary: `benchmark-results/longmemeval-v5-full-500q.json`.

Reproduction:

```
bash scripts/download_longmemeval.sh
bash scripts/run_longmemeval_prod.sh
python scripts/run_longmemeval.py \
  --types knowledge-update --systems acme,rag,memgpt
```

(Azure API for the prod script; full 500 Q: `LONGMEMEVAL_TYPES=all`.)

Results export to `benchmark-results/longmemeval-latest.json`. LongMemEval measures **QA accuracy over chat history**; MemoryBench measures **belief lifecycle and feedback** — the two benchmarks are complementary and must not be merged into a single score.

> **Key finding.** On the full LongMemEval oracle split (500 Q, GPT-4.1, single clean run), ACME v5 hybrid routing reaches **87.6%** overall vs. 77.6% (RAG) and 78.6% (MemGPT) — +10.0 points over RAG — with gains on `temporal-reasoning` (80.3%), abstention (83.3%), and `knowledge-update` (94.4%) while preserving MemoryBench belief/feedback advantages (Section 6).

6. Experimental Setup

Table 7: Experimental configuration (Azure production).

Component	Configuration
LLM	Azure OpenAI GPT-4.1
Embeddings	`text-embedding-3-small`, 256D
Database	Postgres Flexible + pgvector
Graph	Neo4j 5.x on Container Apps
Judge	Same deployment as extractor/reasoner
Reproducibility	`/benchmark/memorybench`, async compare

Model-sensitivity runs additionally use GPT-4.1-mini and GPT-5.4 (same endpoint family; GPT-5.4 requires `max_completion_tokens` API parameter).

Table 4: LongMemEval oracle protocol (industry benchmark [37]).

Component	Configuration
Dataset	`longmemeval_oracle.json` (500 Q, evidence sessions)
Ingest	ACME orchestrator / RAG / MemGPT on identical haystacks
Scoring	Official type-specific yes/no judge (reference `evaluate_qa.py`)
Metrics	Overall accuracy + per `question_type` breakdown
Reproduce	`scripts/run_longmemeval.py` (see `docs/LONGMEMEVAL.md`)

Table 5: LongMemEval `knowledge-update` subset (78 Q, oracle, GPT-4.1, June 2026).

System	Overall	KU	Abst.
ACME (transcript-first)	**0.897**	**0.931**	0.500
MemGPT	0.872	0.903	0.500
RAG	0.859	0.875	0.667

Table 6: LongMemEval oracle full split (500 Q, GPT-4.1, June 2026).

Sys.	All	KU	Multi	Temp.	SS-U	SS-A	SS-P
ACME	**0.876**	**0.944**	**0.793**	**0.803**	**1.000**	**1.000**	**0.900**
MemGPT	0.786	0.861	0.793	0.630	1.000	0.982	0.600
RAG	0.776	0.875	0.793	0.622	1.000	0.964	0.467

7. Results

Run metadata: API revision `acme-api--membenchfix`; compare job `3b31e5e3-72b1-4ce3-b6c2-848cf94e4128`; duration 725 s; zero scenario failures.

Table 8: Primary benchmark (GPT-4.1, 13 scenarios, job `3b31e5e3`). Ret.=Retention, Gnd.=Groundedness, Fbk.=Feedback, Bel.=Belief quality, Ovr.=Overall.

System	Ret.	Gnd.	Fbk.	Bel.	Ovr.
ACME	**1.000**	**1.000**	**1.000**	**0.700**	**0.925**
RAG	0.969	0.977	—	—	0.487
MemGPT	0.977	0.969	—	—	0.487
LangGraph	0.900	0.977	—	—	0.469

Baselines exclude feedback/belief in overall (not applicable). Scores rounded to three decimals from persisted `benchmark_runs` export.

> **Key finding.** ACME's advantage comes primarily from **feedback correction** and **belief quality**, rather than from raw retention alone. ACME gains **+0.44** overall vs RAG on GPT-4.1 while retention and groundedness remain competitive ($\geq$ 0.90) across systems [9,32].

7.1 Model sensitivity (GPT-4.1-mini)

Job `107d02c0`, 639 s. ACME retains **+0.39** overall vs RAG (0.858 vs 0.473). Feedback stays at 1.000 and belief at 0.700 — cognitive layers compensate when the base model is weaker [4,36].

Table 9: Model sensitivity — GPT-4.1-mini (job `107d02c0`).

System	Ret.	Gnd.	Fbk.	Bel.	Ovr.
ACME	**0.892**	**0.838**	**1.000**	**0.700**	**0.858**
RAG	0.908	0.985	—	—	0.473
MemGPT	0.915	1.000	—	—	0.479
LangGraph	0.854	0.862	—	—	0.429

7.2 Model sensitivity (GPT-5.4)

Job `eb68f028`, 620 s. ACME **0.873** vs RAG **0.474** (**+0.40**).

Table 10: Model sensitivity — GPT-5.4 (job `eb68f028`).

System	Ret.	Gnd.	Fbk.	Bel.	Ovr.
ACME	**0.952**	**0.839**	**1.000**	**0.701**	**0.873**
RAG	0.964	0.931	—	—	0.474
MemGPT	0.975	0.939	—	—	0.478
LangGraph	0.973	0.950	—	—	0.481

Belief quality remains **~0.70** across all three LLM tiers — suggesting the external belief substrate [3,21] is comparatively stable while retention/groundedness track model capability [12].

7.3 Component ablations

We disable one cognitive layer at a time via environment flags (`scripts/run_ablation_prod.sh`). Table 2 reports ACME-only MemoryBench v3.1 scores (no baseline compare — 4x cost reduction per ablation).

Full ACME row from 13-scenario compare (job `3b31e5e3`). Ablation rows from ACME-only runs on GPT-4.1 (24 June 2026, premium ingress, stamp `20260624213154`).

> **Key finding.** Disabling belief sync collapses belief quality to **0.000** and overall score by $-$**0.19**, confirming the belief engine as the primary differentiator.

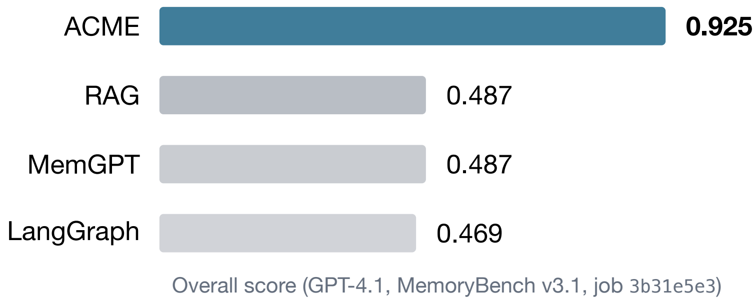

Figure 3: Result summary: ACME leads primarily on feedback and belief-quality dimensions unavailable to baselines.

Table 11: Component ablations (ACME-only, MemoryBench v3.1, GPT-4.1).

Configuration	Ovr.	Ret.	Gnd.	Fbk.	Bel.	Δ
Full ACME	**0.925**	**1.000**	**1.000**	**1.000**	**0.700**	—
No contrarian	0.923	1.000	0.993	1.000	0.700	−0.002
No belief sync	0.731	0.996	1.000	0.929	0.000	−0.194
No vector retrieval	0.923	0.993	1.000	1.000	0.700	−0.002

Contrarian and vector ablations each reduce overall by only −**0.002**.

8. Limitations

Key limitations.

- LLM judge and extractor introduce variance [12,13]; we report sandbox-isolated runs with persisted `benchmark_runs`.
- Scenarios emphasize SaaS churn/latency plus one healthcare transfer set; broader domains and multilingual evals needed [37,38].
- Baselines are reference implementations, not full upstream MemGPT/LangGraph codebases [10,11].
- Production API requires API key for benchmark endpoints.
- CRS weights are hand-tuned; calibration against human epistemic judgments [41] is future work.
- **LongMemEval routing.** The LongMemEval adapter uses per-type answer paths (transcript-first for KU, vector-ranked aggregation for multi-session, abstention and preference prompts); MemoryBench still uses the graph + CRS query path.
- **Multi-session parity.** After v4 routing, ACME matches RAG on `multi-session` (79.3% each on the 241-Q v4 run); dense retrieval remains competitive when evidence is scattered without temporal conflict.
- **Abstention.** Full-split abstention accuracy reaches 83.3% (v5) via anchor short-circuit; harder entity-substitution probes remain imperfect.

9. Conclusion

ACME shows that externalizing belief and learning from LLM weights yields stronger cognitive memory than vector RAG or lightweight agent-memory baselines on MemoryBench, especially for feedback-driven correction and auditable belief quality. On the official LongMemEval oracle split (500 Q, GPT-4.1), ACME v5 hybrid routing reaches **87.6%** overall vs. **77.6%** (RAG), with the largest margins on knowledge-update and temporal-reasoning types. The consistent **+0.39–0.44** margin over RAG on MemoryBench across GPT-4.1, GPT-4.1-mini, and GPT-5.4 supports the hypothesis that explicit belief lifecycle management [21,22,54,55] complements — rather than replaces — advances in base model capability [1,47]. Concurrent belief-memory systems [50,54,56,57] validate the problem direction; ACME contributes reproducible production evidence that epistemic layers — not retrieval alone — drive the measured gap on MemoryBench, while benchmark-specific transcript routing closes industry QA gaps without abandoning the cognitive substrate.

10. References

1. Bommasani, R. et al. On the Opportunities and Risks of Foundation Models. *arXiv:2108.07258*, 2021.
2. McKenna, B. et al. Towards Reliable Generation: Rethinking Faithfulness and Hallucination in LLMs. *Findings of ACL*, 2024.
3. Anderson, J. R., Bothell, D., Byrne, M. D., Douglass, S., Lebiere, C., & Qin, Y. An Integrated Theory of the Mind. *Psychological Review*, 111(4), 1036–1060, 2004.
4. Evans, J. S. B. T. & Stanovich, K. E. Dual-Process Theories of Higher Cognition. *Perspectives on Psychological Science*, 8(3), 223–241, 2013.
5. Gao, L. et al. Retrieval-Augmented Generation for Large Language Models: A Survey. *arXiv:2312.10997*, 2023.
6. Wang, X. et al. Searching for Best Practices in Retrieval-Augmented Generation. *arXiv:2407.01219*, 2024.
7. Madaan, A. et al. Self-Refine: Iterative Refinement with Self-Feedback. *NeurIPS*, 2023.
8. Saunders, W. et al. Self-Critiquing Models for Assisting Human Evaluators. *arXiv:2206.05802*, 2022.
9. Lewis, P. et al. Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. *NeurIPS*, 2020.
10. Packer, C. et al. MemGPT: Towards LLMs as Operating Systems. *arXiv:2310.08560*, 2023.
11. LangChain. LangGraph: Building Stateful, Multi-Actor Applications with LLMs. Documentation, 2024. https://langchain-ai.github.io/langgraph/
12. Zheng, L. et al. Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena. *NeurIPS*, 2023.
13. Liu, Y. et al. G-Eval: NLG Evaluation using GPT-4 with Better Human Alignment. *EMNLP*, 2023.
14. Karpukhin, V. et al. Dense Passage Retrieval for Open-Domain Question Answering. *EMNLP*, 2020.
15. Khattab, O. & Zaharia, M. ColBERT: Efficient and Effective Passage Search via Contextualized Late Interaction. *SIGIR*, 2020.
16. Park, J. S. et al. Generative Agents: Interactive Simulacra of Human Behavior. *UIST*, 2023.
17. Laird, J. E. The SOAR Cognitive Architecture. *MIT Press*, 2012.
18. Tulving, E. Episodic and Semantic Memory. In *Organization of Memory*, 1972.
19. Schacter, D. L. & Tulving, E. Memory Systems. *MIT Press*, 1994.
20. Rasmussen, D. et al. Zep: A Temporal Knowledge Graph Architecture for Agent Memory. *arXiv:2501.13956*, 2025.
21. Doyle, J. A Truth Maintenance System. *Artificial Intelligence*, 12(3), 231–272, 1979.
22. Gardenfors, P. *Knowledge in Flux: Modeling the Dynamics of Epistemic States*. MIT Press, 1988.
23. pgvector. Open-source vector similarity search for PostgreSQL. https://github.com/pgvector/pgvector
24. Neo4j, Inc. Neo4j Graph Database. https://neo4j.com/
25. Parisi, G. I. et al. Continual Lifelong Learning with Neural Networks: A Review. *Neural Networks*, 113, 54–71, 2019.
26. Kemper, N. & Jankowski, S. Forgetting in Deep Learning — A Survey. *arXiv:2307.09218*, 2023.
27. Ma, X. et al. Query Rewriting for Retrieval-Augmented Large Language Models. *EMNLP*, 2023.
28. Asai, A. et al. Self-RAG: Learning to Retrieve, Generate, and Critique through Self-Reflection. *ICLR*, 2024.
29. Yan, S. et al. Corrective Retrieval Augmented Generation. *arXiv:2401.15884*, 2024.
30. Edge, D. et al. From Local to Global: A Graph RAG Approach to Query-Focused Summarization. *arXiv:2404.16130*, 2024 (Microsoft GraphRAG).
31. Wang, M. et al. Knowledge Graph Prompting for Multi-Document Question Answering. *EMNLP*, 2023.
32. Edge, D. et al. GraphRAG: Unlocking LLM Discovery on Narrative Private Data. Microsoft Research, 2024.
33. Shinn, N. et al. Reflexion: Language Agents with Verbal Reinforcement Learning. *NeurIPS*, 2023.
34. Zhao, A. et al. ExpeL: LLM Agents Are Experiential Learners. *AAAI*, 2024.
35. Bai, Y. et al. Constitutional AI: Harmlessness from AI Feedback. *arXiv:2212.08073*, 2022.
36. McClelland, J. L., McNaughton, B. L., & O'Reilly, R. C. Why There Are Complementary Learning Systems in the Hippocampus and Neocortex. *Psychological Review*, 102(3), 419–457, 1995.
37. Wu, Y. et al. LongMemEval: Benchmarking Long-Term Memory in LLMs. *arXiv:2410.10813*, 2024.
38. Maharana, D. et al. Evaluating Very Long-Term Conversational Memory of LLM Agents (LoCoMo). *ACL*, 2024.
39. Zhong, W. et al. MemoryBank: Enhancing Large Language Models with Long-Term Memory. *AAAI*, 2024.
40. Wang, X. et al. Text2KG: A Survey on Knowledge Graph Construction from Text. *arXiv:2404.09425*, 2024.
41. Guo, C. et al. On Calibration of Modern Neural Networks. *ICML*, 2017.
42. Kadavath, S. et al. Language Models (Mostly) Know What They Know. *arXiv:2207.05221*, 2022.
43. Pearl, J. *Causality: Models, Reasoning, and Inference*. Cambridge University Press, 2009.
44. Halpern, J. Y. *Actual Causality*. MIT Press, 2016.
45. Ji, Z. et al. Survey of Hallucination in Natural Language Generation. *ACM Computing Surveys*, 55(12), 2023.

46. Min, S. et al. FactScore: Fine-grained Atomic Evaluation of Factual Precision in Long Form Text Generation. *EMNLP*, 2023.
47. OpenAI. GPT-4.1 Model Card and System Documentation, 2025.
48. Neelakantan, A. et al. Text and Code Embeddings by Contrastive Pre-Training. *arXiv:2201.10005*, 2022.
49. Kamil Bourouiba. ACME: Adaptive Cognitive Memory Engine (code and benchmarks). https://github.com/KamilBourouiba/ACME 2026.
50. Latif, E. et al. Hindsight is 20/20: Building Agent Memory that Retains, Recalls, and Reflects. *arXiv:2512.12818*, 2025.
51. Chen, Y. et al. Reflection-Bench: Evaluating Epistemic Agency in Large Language Models. *arXiv:2410.16270*, 2024.
52. MemoryAgentBench authors. Evaluating Memory in LLM Agents via Incremental Multi-Turn Interactions. *arXiv:2507.05257*, 2025.
53. Ai, J. et al. MemBench: Towards More Comprehensive Evaluation on the Memory of LLM-based Agents. *arXiv:2506.21605*, 2025.
54. TruthKeeper authors. TruthKeeper: A Curated Memory Architecture for LLMs with Dependency-Aware Truth Maintenance. *arXiv preprint*, 2024.
55. MnemeBrain. Belief-State Memory for AI Agents. https://github.com/mnemebrain/mnemebrain-lite, 2025.
56. NeuSymMS authors. NeuSymMS: A Hybrid Neuro-Symbolic Memory System for LLM Agents. *arXiv:2605.17596*, 2026.
57. Graph-Native Cognitive Memory authors. Formal Belief Revision Semantics for Versioned Memory Architectures. *arXiv:2603.17244*, 2026.
58. Alchourron, C. E., Gardenfors, P., & Makinson, D. On the Logic of Theory Change: Partial Meet Contraction and Revision Functions. *Journal of Symbolic Logic*, 50(2), 510–530, 1985.

11. Appendix A: Reproduction

```
git clone https://github.com/KamilBourouiba/ACME.git && cd ACME
pip install -e ".[dev]"
pytest tests/test_benchmark_gate.py tests/test_memorybench.py -q
# Production (requires API key):
./azure/configure-premium-ingress.sh
./scripts/run_prod_benchmark.sh
./scripts/run_model_benchmark.sh gpt-4.1-mini   # optional model sweep
./scripts/run_ablation_prod.sh                # component ablations
```

12. Appendix B: Data Availability

Benchmark payloads are stored in PostgreSQL table `benchmark_runs` (columns: `overall_score`, `retention_score`, `belief_quality_score`, `payload JSONB`, `created_at`). Export via `GET /api/v1/benchmark/export`. Full run tables are documented in `docs/BENCHMARK_RESULTS.md`~[4